

1) (b) Data and a pointer to the next node.

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2) (c) Null

3) (b) Self referention structure

4) (b) $head = null$

5) (b) $newnode \rightarrow next = head; head = newnode$

6) (b) $temp = head; head = head \rightarrow next; free(temp)$

7) (c) Traversing from tail to head

8) (b) n

9) (c) Copying the data of $P \rightarrow next$ into P , then deleting $P \rightarrow next$

10) (b) $while(P \neq null)$

11) (b) $P == head$

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12) (c) Two pointers advancing at different rates

13) (c) Three

14) (b) Exactly two nodes

15) (b) $P = (struct\ node*) malloc(Sizeof(str))$

16) (b) Avoid special-case handling for insertion and deletion at the head.

17) (c) The first node

18) (b) it does not support random access

19) (b) Loss of access to all nodes, causing a memory leak

20) (b) A slow pointer and a fast pointer

21) (c) The last node

22) (b) Nodes may be scattered anywhere in memory

23)

```
struct node {  
    int data;  
    struct node* next;  
};
```

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24) 30, 50, 70

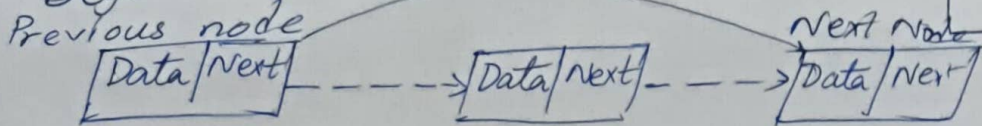
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```
25) void insertEnd (struct node**  
                      head, int x) {  
    struct node t = (struct node) malloc  
        (sizeof(struct node));  
    t->data = x;  
    t->next = null;  
    if (*head == null) {  
        *head = t;  
        return;  
    }  
    struct node *p = *head;  
    while (p->next != Null)  
        p = p->next;  
    p->next = t;  
}
```

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```
26) struct node *reverse  
    (struct node *head) {  
    struct node *prev = Null;  
    struct node *curr = head;  
    struct node *next = null;  
    while (curr != null) {  
        next = curr->next;  
        curr->next = prev;  
        prev = curr;  
        curr = next;  
    }  
    return prev;  
}
```


(27) Before



Pointer adjustment

prev $\rightarrow$ next = del $\rightarrow$ next,
free(del);

(28) 5 $\rightarrow$ 8 $\rightarrow$ 12 $\rightarrow$ 3 $\rightarrow$ Null

7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 12 $\rightarrow$ 3 $\rightarrow$ Null

7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ 9 $\rightarrow$ Null

Final $\rightarrow$ 7 $\rightarrow$ 5 $\rightarrow$ 8 $\rightarrow$ 3 $\rightarrow$ 9 $\rightarrow$ Null

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